

An Analysis of Cross entropy loss

Cross entropy loss

$$H(p, q) = -\sum_x p(x) \log q(x)$$

Cross entropy loss -- Part 1

where N is the size of the test set, and $q(x)$ is the probability of event x estimated from the training set. In other words, $q(x_i)$ is the probability estimate of the model that the i -th word of the text is x_i . The sum is averaged over the N words of the test. This is a Monte Carlo estimate of the true cross-entropy, where the test set is treated as samples from $p(x)$.

Cross entropy loss -- Part 2

where $E_p[\cdot]$ is the expected value operator with respect to the distribution p . The definition may be formulated using the Kullback–Leibler divergence $D_{KL}(p \parallel q)$, divergence of p from q (also known as the relative entropy of p with respect to q).

Cross entropy loss -- Part 3

In information theory, the cross-entropy between two probability distributions p and q , over the same underlying set of events, measures the average number of bits needed to identify an event drawn from the set when the coding scheme used for the set is optimized for an estimated probability distribution q , rather than the true distribution p .

Cross entropy loss -- Part 4

Relation to maximum likelihood The cross entropy arises in classification problems when introducing a logarithm in the guise of the log-likelihood function. This section concerns the estimation of the probabilities of different discrete outcomes. To this end, denote a parametrized family of distributions by q_θ , with θ subject to the optimization effort. Consider a given finite sequence of N values x_i from a training set, obtained from conditionally independent sampling. The likelihood assigned to any considered parameter θ of the model is then given by the product over all probabilities $q_\theta(X=x_i)$

. Repeated occurrences are possible, leading to equal factors in the product. If the count of occurrences of the value equal to x is denoted by $\#x$, then the frequency of that value equals $\#x/N$. If $p(X=x)$ is the underlying probability distribution, for large N we expect $p(X=x) \approx \#x/N$, by the law of large numbers. Writing our likelihood function as the product of observations from the distribution q_θ :

Cross entropy loss -- Part 5

Since the logarithm is a monotonically increasing function, the maximizing value of θ is unaffected by this final step. Similarly, the maximizing value of θ is unaffected by the factor of N . So we observe that the likelihood maximization amounts to minimization of the cross-entropy.

Cross entropy loss -- Part 6

$$\frac{\partial}{\partial w_0} \ln \left(\frac{1}{1 + e^{-w_0 + k_0}} \right) = \frac{\partial}{\partial w_0} \ln \left(\frac{1}{1 + e^{-w_0 + k_0}} \right) = -\frac{1}{1 + e^{-w_0 + k_0}} = -\frac{y}{1 + e^{-w_0 + k_0}} = -\frac{y}{1 + e^{-w_0 + k_0}}$$

Cross entropy loss -- Part 7

$$q_{y=1} = \frac{1}{1 + e^{-w \cdot x}} = \frac{1}{1 + e^{-w \cdot x}}, \quad q_{y=0} = \frac{1}{1 + e^{-w \cdot x}}$$

Cross entropy loss -- Part 8

$$\frac{\partial}{\partial w_0} \ln \left(\frac{1}{1 + e^{-w_0 + k_0}} \right) = \frac{\partial}{\partial w_0} \ln \left(\frac{1}{1 + e^{-w_0 + k_0}} \right) = -\frac{1}{1 + e^{-w_0 + k_0}} = -\frac{y}{1 + e^{-w_0 + k_0}}$$

Cross entropy loss -- Part 9

Amended cross-entropy It may be beneficial to train an ensemble of models that have diversity, such that when they are combined, their predictive accuracy is augmented. Assuming a simple ensemble of K classifiers is assembled via averaging the outputs, then the amended cross-entropy is given by

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Cross entropy loss -- Part 10

Relation to linear regression The gradient of the cross-entropy loss for logistic regression is equal to the gradient of the squared-error loss for linear regression (up to a constant factor). To see this, define

Cross entropy loss -- Part 11

Cross-entropy loss function and logistic regression Cross-entropy can be used to define a loss function in machine learning and optimization. Mao, Mohri, and Zhong (2023) give an extensive analysis of the properties of the family of cross-entropy loss functions in machine learning, including theoretical learning guarantees and extensions to adversarial learning. The true probability p_i is the true label, and the given distribution q_i is the predicted value of the current model. This is also known as the log loss (or logarithmic loss or logistic loss); the terms "log loss" and "cross-entropy loss" are used interchangeably. More specifically, consider a binary regression model which can be used to classify observations into two possible classes (often simply labelled 0 and 1). The output of the model for a given observation, given a vector of input features x , can be interpreted as a probability, which serves as the basis for classifying the observation. In logistic regression, the probability is modeled using the logistic function $g(z) = 1 / (1 + e^{-z})$ where z is some function of the input vector x , commonly just a linear function. The probability of the output $y = 1$ is given by

Cross entropy loss -- Part 12

where e^k is the cost function of the k^{th} classifier, q_k is the output probability of the k^{th} classifier, p is the true probability to be estimated, and λ is a parameter between 0 and 1 that defines the 'diversity' that we would like to establish among the ensemble. When $\lambda = 0$ we want each classifier to do its best regardless of the ensemble and when $\lambda = 1$ we would like the classifier to be as diverse as possible.